

Molecular Shapes (VSEPR)

Answer key with worked solutions

A. Understanding

1. VSEPR.

Electron domains (bonds and lone pairs) around a central atom repel each other and arrange themselves **as far apart as possible**, which sets the shape of the molecule.

2. Domain geometry vs shape.

The electron-domain geometry describes **all** domains, including lone pairs; the molecular shape describes only the positions of the **atoms**. Water has 4 domains (tetrahedral electron-domain geometry), but 2 are lone pairs, so its molecular shape is **bent**.

B. Predicting shapes

3. Shapes.

(a) BCl_3 : 3 domains, 0 lone pairs, **trigonal planar**, 120° (b) H_2S : 4 domains, 2 lone pairs, **bent**, less than 109.5° (c) PCl_3 : 4 domains, 1 lone pair, **trigonal pyramidal**, less than 109.5° (d) CS_2 : 2 domains (two $\text{C}=\text{S}$), 0 lone pairs, **linear**, 180° (e) SiH_4 : 4 domains, 0 lone pairs, **tetrahedral**, 109.5°

4. Formaldehyde.

Carbon has 3 domains (two $\text{C}-\text{H}$ bonds and one $\text{C}=\text{O}$ double bond, counted once) and no lone pairs: **trigonal planar**, $\text{H}-\text{C}-\text{H}$ about 120° .

5. Ammonium.

N has 4 bonding domains and no lone pairs: **tetrahedral**, 109.5° .

C. Bond angles

6. Trend.

All three have 4 electron domains. Lone pairs repel **more strongly** than bonding pairs, so they push the bonds closer together. CH_4 has 0 lone pairs, NH_3 has 1 and H_2O has 2, so the angle decreases as the number of lone pairs increases.

D. Polarity

7. Polar or nonpolar.

(a) CO_2 : **nonpolar**; linear, so the two $\text{C}=\text{O}$ polarities cancel (b) H_2O : **polar**; bent, so the $\text{O}-\text{H}$ polarities don't cancel (c) CCl_4 : **nonpolar**; tetrahedral with four identical bonds, so they cancel (d) NH_3 : **polar**; trigonal pyramidal, so the $\text{N}-\text{H}$ polarities don't cancel

E. Challenge

8. XeF_4 .

Six domains point to the corners of an octahedron. The two lone pairs go **opposite each other** (180° apart) to minimise lone pair-lone pair repulsion, leaving the four F atoms in a square: **square planar**, $\text{F}-\text{Xe}-\text{F}$ 90° .

9. SF_4 .

Five domains with one lone pair, which sits in an equatorial position: **seesaw** shape.